8.4 EXAMPLES

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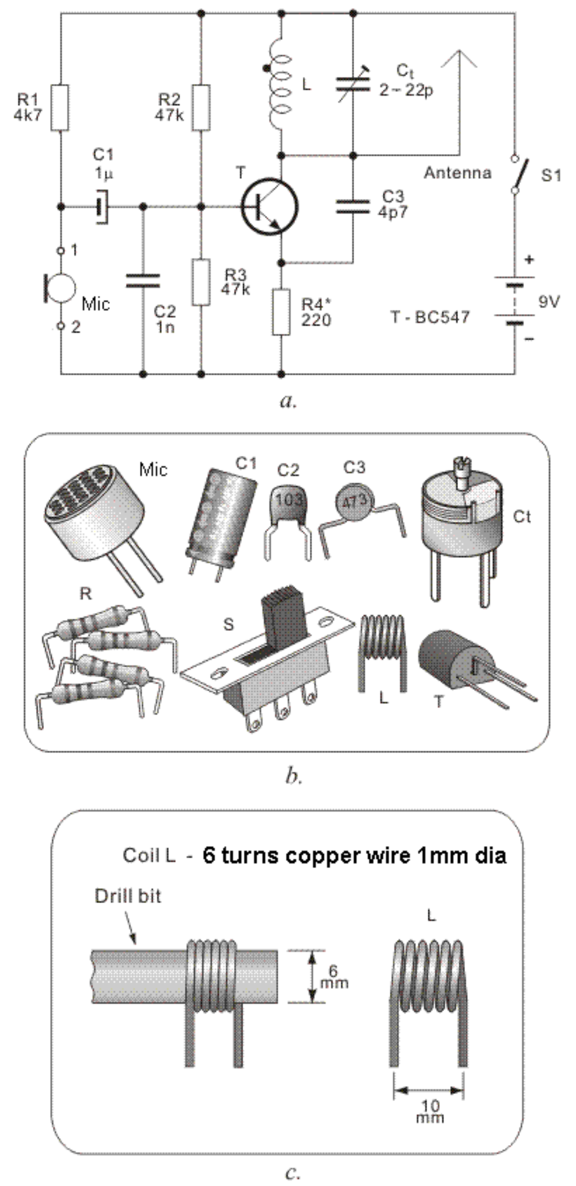
The schematic of a very simple FM radio-transmitter is shown in figure 8.6. It uses an electret microphone and transmits on a frequency between 88MHz and 108MHz.

The transistor, coil L, trimmer capacitor Ct, capacitor C3 and resistors R2, R3 and R4 creates an oscillator with a frequency determined by:

$$f_o = \frac{1}{2\pi\sqrt{L(C_t + C_{CB})}}.$$

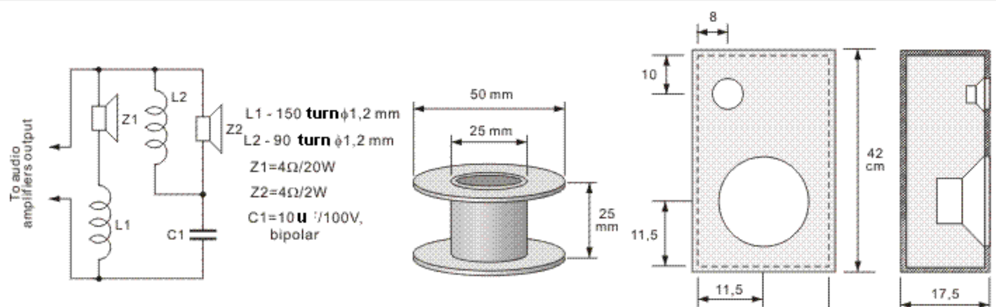
In this equation C_{CB} represents the capacitance between the collector and the base. The value of this capacitance depends on the voltage on the base. The higher the voltage, the lower the capacitance and vice versa. The voltage on the base is constant while there is no sound, which means the frequency of the oscillator is constant. When the microphone picks up a sound, it is passed to the base of the transistor via C1. This causes the frequency of the oscillator to change and that’s why the circuit is called FREQUENCY MODULATED (FM).
To transmit on a frequency away from any other radio station, a trim cap is included. The transmitter has a range up to 200 metres, depending on the length of the antenna and where it is placed. Ideally, the antenna should be vertical and as high as possible.
The antenna can be as long as 3 metres but 180cm will work very well.
Coil L is made by winding 6 turns of 1mm enameled wire on a 6mm dia drill bit. This coil can be stretched of squashed to adjust the operating frequency of the circuit and the trimmer will fine tune the frequency.



Sl. 8.6. Simple FM radio transmitter:
a - schematic, b - components, c -making the coil

High Fidelity (or Hi-Fi) sound reproduction is the main purpose for using good-quality speakers. They are used in radios, TV's, cassette players, CD players, etc. The speakers are housed in speaker boxes and use at least two speakers. This is because no individual speaker is capable of reproducing the full range of frequencies. A speaker with a large cone is called a "Woofer" and will reproduce the low frequencies. A speaker with a small cone is called a "tweeter" and will reproduce the high frequencies. Together, they will reproduce the full range of between 30Hz and 15kHz.

The difficulty is now to detect the low or high frequency and divert the correct frequency to the particular speaker. This is the job of a cross-over network. In the figure 8.7 an inductor L1 passes the low frequencies to speaker Z1 and capacitor C1 passes the high frequencies to speaker Z2. Z1 reproduces frequencies from 30Hz to 800Hz and Z2 reproduces sounds with frequencies from 800Hz to 15kHz.

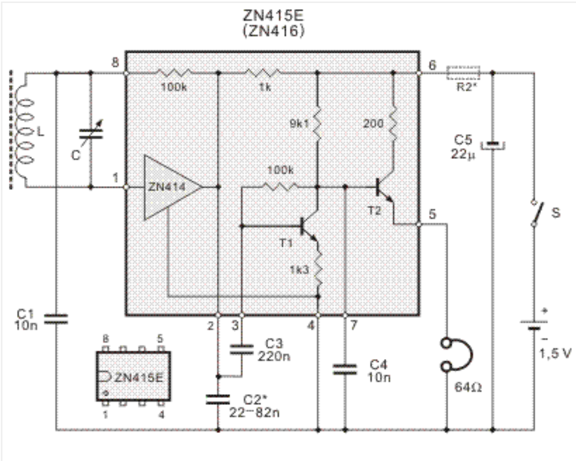


8.7. **Speaker box**

Headphones are most commonly used with portable devices, such as radio receivers, cassette players, CD walkmans, mp3 players, etc. Headphones produce a very high quality reproduction. All modern devices have an audio-amplifier. It usually employs an integrated circuit and most of these are designed for 32 ohm headphones. There are also 8 ohm and 16 ohm headphones.

The schematic of a AM portable radio is shown in figure 8.8. It’s built around the ZN416 integrated circuit. The output is connected to two serially connected 32 ohm headphones, with overall resistance of 64 ohms.

It is possible to connect the radio receiver in figure 8.8 to amplifier in figure 7.3 to produce a radio with speaker output.



8.8. Mini AM radio-receiver using the ZN415E